Two-Sample Student-t-Tests

05 November 2026 22:02
Based off of 9ST0 Content
lbweb.uk/publications

Two-Sample t-Tests are parametric statistical tests in which determines if the mean of two independent groups differ significantly.
This is carried out by dividing the difference in mean by the standard error to produce a test statistic that is then tested against the critical value or region.

A two-Sample t-Test is used when,
a) The two samples are independent of one another (not paired data)
b) Variance of the <u>population</u> is unknown
c) Both populations are normally distributed
Using this test, we must assume that the population variances are equal to one another

Formulae:

Pooled Variance, $$s_p^2 = \frac{(n_x - 1)\cdot s_x^2 + (n_y - 1)\cdot s_y^2}{n_x + n_y - 2}$$

Where, "s_x^2" and "s_y^2" are the variances of each sample, x and y
- "$n_x + n_y - 2$" are the degrees of freedom

Test Statistic, $$t_{n_x+n_y-2} = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{\sqrt{s_p^2 \cdot \left(\frac{1}{n_x} + \frac{1}{n_y}\right)}}$$

where, "$(\bar{x} - \bar{y})$" is the difference of sample means.
- "$(\mu_x - \mu_y)$" is the <u>hypothesised</u> difference
  - e.g. if "$H_1: \mu_x - \mu_y > 5$", then were testing if the difference in population mean is greater than 5.
  With this, "$(\mu_x - \mu_y)$" would become "5"
- "$\sqrt{s_p^2 \cdot \left(\frac{1}{n_x} + \frac{1}{n_y}\right)}$" is the standard error

Degrees of Freedom, $df = n_x + n_y - 2$

Example no. 1

1. Two groups of students take the same examination. A random sample of students is taken from each of the groups. The marks of the 9 students from Group 1 are as follows

μ_1 30 29 35 27 23 33 33 35 28

The marks, y, of the 7 students from Group 2 gave the following statistics

μ_2 $\bar{y} = 31.29, \quad s_y^2 = 12.9.$

A test is to be carried out to see whether or not there is a difference between the mean marks of the two groups of students. You may assume that the samples are taken from normally distributed populations and that they are independent.

(a) State one other assumption that must be made in order to apply this test.

(b) Using this assumption, test whether or not there is a difference between the mean marks of the two groups of students.

Source: Two-Sample t-Tests : Content

a) The population variences must be equal to one another

← not specified, so default to $\alpha = 0.05$

b) Two-Sample t-Test at 5% Significance

$H_0: \mu_1 - \mu_2 = 0$ where μ_1 represents the population mean of marks students from group 1 scored on a test

$H_1: \mu_1 - \mu_2 \neq 0$ μ_2 represents the population mean of marks students from group 2 scored on a test

input data into CG50's stats mode.

μ_1	μ_2
$\bar{x}_1 = 30.333$	$\bar{x}_2 = 31.29$
$S_1^2 = 16.24999$	$S_2^2 = 12.9$
$n_1 = 9$	$n_2 = 7$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{(9-1)\cdot 16.24999 + (7-1)\cdot 12.9}{9+7-2} = 14.81428$$

← No hypothesised difference

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{(30.333 - 31.29)}{\sqrt{14.81428\left(\frac{1}{9}+\frac{1}{7}\right)}} = -0.4932$$

$df = n_1 + n_2 - 2 = 9+7-2 = 14$

$\alpha = 0.05 \div 2 = 0.025$ ← two-tailed as $\neq$

$\therefore CV = 2.145$ → Table 5 p.g. 19

p	0.9	0.95	0.975	...
ν				...
1	3.078	6.314	12.706	
2	1.886	2.920	4.303	
3	1.638	2.353	3.182	
4	1.533	2.132	2.776	
5	1.476	2.015	2.571	
6	1.440	1.943	2.447	
7	1.415	1.895	2.365	
8	1.397	1.860	2.306	
9	1.383	1.833	2.262	
10	1.372	1.812	2.228	
11	1.363	1.796	2.201	
12	1.356	1.782	2.179	
13	1.350	1.771	2.160	
14	1.345	1.761	2.145	

$|TS| < CV$

$0.4932 < 2.145$

$\therefore$ Fail to reject H_0

There is insufficient evidence at 5% Significance to suggest that group 1's mean marks where different to that of group 2's mean marks.

Example no. 2

2. A biologist investigating the shell size of turtles takes random samples of adult female and adult male turtles and records the length, in cm, of the shell. The results are as follows:
   - From a sample of 6 adult female turtles, the mean length is 19.6 and the sample variance is 0.61.
   - From a sample of 12 adult male turtles, the mean length is 13.7 and the sample variance is 0.935.

   You may assume that samples come from independent normal distributions with the same variance.
   The biologist claims that the mean shell length of adult female turtles is 5 cm longer than the mean shell length of adult male turtles. Test the biologist's claim.

hypothesised difference

Two-Sample t-Test at a 5% Significance Level

$H_0: \mu_f - \mu_m = 5$ where μ_f represents the population mean length of female adult turtle's shells in cm

$H_1: \mu_f - \mu_m > 5$ (hypothesised difference) μ_m represents the population mean length of male adult turtle's shells in cm

μ_f	μ_m
$\bar{x}_f = 19.6$	$\bar{x}_m = 13.7$
$s_f^2 = 0.61$	$s_m^2 = 0.935$
$n_f = 6$	$n_m = 12$

$$s_p^2 = \frac{(n_f - 1)s_f^2 + (n_m - 1)s_m^2}{n_f + n_m - 2} = \frac{(6-1)\cdot 0.61 + (12-1)\cdot 0.935}{6 + 12 - 2} = 0.8334375$$

hypothesised difference of 5

$$t = \frac{(\bar{x}_f - \bar{x}_m) - 5}{\sqrt{s_p^2\left(\frac{1}{n_f} + \frac{1}{n_m}\right)}} = \frac{(19.6 - 13.7) - 5}{\sqrt{0.8334375\left(\frac{1}{6} + \frac{1}{12}\right)}} = 1.972$$

$df = 6 + 12 - 2 = 16$
$\alpha = 0.05$
$\therefore CV = 1.746$

TS > CV
1.972 > 1.746
∴ Fail to accept H_0

There is sufficient evidence at a 5% Significance level to suggest that the length of turtle's shells are 5cm greater in female adult turtles than male adult turtles.